

SIRDAL-SINNES, Norway

WMO: 014310

Lat: 58.9230N

Lon: 6.9096E

Elev: 1827

StdP: 13.75

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.0	-2.4	-12.2	3.0	-7.6	-6.8	4.1	-1.4	19.0	26.1	17.5	27.0	2.3	30	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	73.7	58.3	70.5	56.8	66.6	55.4	60.5	69.7	58.8	66.9	56.9	64.3	6.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.9	74.0	63.4	55.2	69.4	61.2	53.6	65.3	59.5	27.5	70.0	26.3	66.7	25.1	64.2	65.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	13.2	11.5	-13.5	78.4	4.7	3.7	-16.8	81.0	-19.6	83.2	-22.2	85.3	-25.6	88.0
			-12.9	62.8	4.5	2.0	-16.2	64.2	-18.8	65.4	-21.4	66.5	-24.6	68.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.7	23.5	23.6	29.3	36.4	44.6	50.7	55.8	53.3	48.3	40.1	32.1	26.1
	DBStd	13.38	9.62	10.11	7.57	6.18	6.90	4.89	5.30	3.94	4.65	5.51	6.93	10.43
	HDD50	4553	823	740	643	411	201	47	8	11	84	308	536	741
	HDD65	9591	1288	1160	1108	859	634	429	288	362	500	771	986	1206
	CDD50	438	0	0	0	1	32	68	187	114	34	2	0	0
	CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0
	CDH74	75	0	0	0	0	4	8	62	1	0	0	0	0
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0
	WSAvg	5.2	5.7	5.6	5.8	5.0	5.3	5.4	4.8	4.6	4.9	5.2	5.1	5.1
	PrecAvg	77.4	9.9	7.2	5.6	4.0	3.5	3.9	4.6	6.2	5.9	9.0	8.4	9.2
(15) Precipitation	PrecMax	112.4	20.6	19.1	16.4	7.5	7.7	7.2	11.0	10.6	11.7	18.3	19.4	20.9
	PrecMin	41.8	0.8	0.7	0.2	1.7	0.4	0.6	1.3	0.9	1.3	1.8	1.1	0.0
	PrecStd	15.5	5.4	5.0	4.0	1.5	2.0	1.9	2.1	2.1	2.7	4.4	4.6	6.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	42.8	50.4	59.9	73.6	74.9	79.2	72.6	67.9	57.8	49.0	43.9
		MCWB	38.2	38.6	41.6	46.7	55.0	56.5	61.0	59.3	55.3	49.9	45.6	41.8
	2%	DB	38.5	38.8	43.3	55.6	69.0	71.1	75.3	67.5	63.0	53.5	45.9	41.3
		MCWB	36.5	35.9	38.2	45.1	53.9	55.3	59.6	57.1	52.9	48.2	43.7	39.6
	5%	DB	36.7	37.1	40.2	50.9	63.7	66.8	72.2	64.5	58.9	50.8	43.3	39.1
		MCWB	34.9	34.7	36.4	42.8	51.0	53.8	58.9	55.2	52.6	47.4	41.3	37.3
	10%	DB	34.9	35.5	38.4	46.7	59.0	62.3	68.3	61.6	56.3	48.5	40.8	37.2
		MCWB	33.2	33.4	34.9	40.2	48.6	51.3	57.0	54.5	51.1	45.5	38.6	35.2
	0.4%	WB	38.7	39.5	42.3	48.4	57.5	59.7	63.6	61.1	58.2	54.4	46.5	41.9
		MCDB	40.3	42.1	48.1	57.0	70.3	69.6	74.6	67.8	64.3	56.4	48.6	42.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.0	36.6	39.1	45.8	55.0	57.1	61.5	59.2	55.2	50.1	43.9	39.4
		MCDB	38.3	38.3	42.4	54.0	66.4	67.2	71.6	65.2	60.0	51.9	45.9	40.8
	5%	WB	35.1	35.1	37.2	43.5	52.1	54.9	59.9	57.1	53.2	48.0	41.6	37.1
		MCDB	36.4	36.7	39.7	50.7	61.1	64.2	69.3	62.4	57.7	50.0	43.4	38.5
	10%	WB	33.3	33.7	35.4	40.8	49.6	52.8	58.2	55.4	51.5	45.6	38.9	35.2
		MCDB	34.7	35.2	37.8	45.9	58.2	60.9	66.5	59.8	55.9	48.2	40.6	36.6
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	14.4	14.1	15.3	17.1	17.1	16.4	14.4	12.7	11.4	9.6	10.9
		MCDBR	9.7	12.0	14.7	23.7	27.1	27.1	25.4	20.0	18.3	15.8	11.0	9.0
		MCWBR	8.9	10.1	10.3	14.6	13.9	13.1	11.8	10.5	10.7	11.1	9.8	8.5
	5% WB	MCDBR	9.6	9.9	12.7	21.9	24.7	22.5	22.4	15.4	14.7	10.7	10.4	9.0
		MCWBR	8.8	8.6	9.4	13.5	13.2	11.9	11.3	9.5	10.1	9.5	9.5	8.6
(40) Clear-Sky Solar Irradiance	taub		0.246	0.278	0.306	0.342	0.343	0.334	0.349	0.346	0.322	0.305	0.265	0.226
	taud		2.168	2.252	2.251	2.278	2.428	2.477	2.445	2.448	2.517	2.486	2.302	2.112
	Ebn at Noon		184	238	265	273	282	286	278	271	261	230	179	148
	Edh at Noon		14	22	30	35	32	31	32	30	23	18	13	10
(44) All-Sky Solar Radiation	RadAvg		106	305	718	1226	1529	1697	1511	1162	750	364	140	71
	RadStd		23	66	114	184	232	185	192	100	82	64	28	10

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-209	N/A	N/A

Nomenclature: See separate page